

1 **Highly diverse endophytic and soil *Fusarium oxysporum* populations associated with field-**
2 **grown tomato plants**

3

4 Jill E. Demers, Beth K. Gugino, María del Mar Jiménez-Gasco,#

5 Department of Plant Pathology and Environmental Microbiology, The Pennsylvania State

6 University, University Park, PA, USA.

7

8 # Address correspondence to María del Mar Jiménez-Gasco, jimenez-gasco@psu.edu

9

10 Running title: Endophytic and soil populations of *Fusarium oxysporum*

11

12 **Abstract**

13 The diversity and genetic differentiation of populations of *Fusarium oxysporum* associated with
14 tomato fields, both endophytes obtained from tomato plants as well as isolates obtained from
15 soil surrounding the sampled plants was investigated. A total of 609 isolates of *F. oxysporum*
16 were obtained, 295 isolates obtained from a total of 32 asymptomatic tomato plants in two
17 fields and 314 isolates from eight soil cores sampled from the area surrounding the plants.
18 Included in this total were 112 isolates from the stems of all 32 plants, a niche that has not
19 been previously included in *F. oxysporum* population genetics studies. Isolates were
20 characterized using the DNA sequence of the translation elongation factor 1- α gene. A diverse
21 population of twenty-six sequence types was found, although two sequence types represented
22 nearly two-thirds of the isolates studied. Sequence types were placed in different phylogenetic
23 clades within *F. oxysporum*, and endophytic isolates were not monophyletic. Multiple sequence
24 types were found in all plants, with an average of 4.2 per plant. Population composition
25 differed between the two fields, but not between soil samples within each field. A certain
26 degree of differentiation was observed between populations associated with different tomato
27 cultivars, suggesting that host genotype may affect the composition of plant-associated *F.*
28 *oxysporum* populations. No clear patterns of genetic differentiation were observed between
29 endophyte populations and soil populations, suggesting a lack of specialization of endophytic
30 isolates.

31

32 Introduction

33 *Fusarium oxysporum* is a well-known phytopathogenic fungus that affects hundreds of
34 crops worldwide. Most studies of this fungus focus on the ability of *F. oxysporum* to cause
35 vascular wilt on a particular host. However, environmental surveys of a wide range of habitats
36 frequently find *F. oxysporum* in the absence of plant disease (1). These *F. oxysporum*
37 populations, usually referred to as nonpathogenic, although lack of pathogenicity is usually not
38 confirmed, are a cosmopolitan component of soil communities. *F. oxysporum* can also infect
39 and colonize plants asymptotically as an endophyte. This endophytic lifestyle has not been
40 well researched, despite reports of endophytic colonization by *F. oxysporum* on nearly 100
41 plant species (2).

42 The interaction of endophytic *F. oxysporum* and plants may give insight into how
43 pathogen recognition and resistance operate. In susceptible plants, pathogenic *F. oxysporum*
44 penetrate the root, colonize the root cortex, and then spread through the xylem to the rest of
45 the plant, causing occlusion of the xylem and ultimately wilting and death (3, 4). In host
46 cultivars that are resistant to the pathogen, the spread of the fungus is blocked, and only
47 limited colonization of host tissue occurs (5, 6, 7). Nonpathogenic *F. oxysporum* inoculated onto
48 plants can colonize the root cortex similar to pathogens, but host responses such as thickening
49 of cell walls and protuberances into intercellular spaces seem to limit the growth of
50 endophytes, similar to defenses limiting the growth of pathogens in resistant plants (3, 4, 8).
51 Nonpathogenic *F. oxysporum* can also have a differential ability to colonize and multiply in the
52 root cortex of different plant species. For example, research done by Leoni et al. (9) showed

53 how *F. oxysporum* f. sp. *cepae* (that causes Fusarium basal rot of onions) can reach high
54 population levels in the roots of plants such as black bean or sudan grass, in the absence of
55 disease. Although the above observations suggest that endophytic *F. oxysporum* are not
56 systemic, *F. oxysporum* endophytes can be recovered from naturally-colonized plant stems (10).
57 However, most population genetics studies have focused on nonpathogenic *F. oxysporum*
58 isolated from roots (11, 12, 13, 14, 15).

59 Despite the cosmopolitan nature of *F. oxysporum* and its ecological diversity, the
60 structure and diversity of soil and endophytic *F. oxysporum* populations are not well
61 understood. A recent typing scheme proposed by O'Donnell et al. (16) used DNA sequences of
62 the ribosomal DNA intergenic spacer region (IGS) and the translation elongation factor 1 α gene
63 (TEF) to define sequence types and found 256 sequence types among 850 isolates, mainly plant
64 pathogens. Subsequent studies using the same typing scheme found 26 new sequence types
65 among isolates obtained from soil in Sardinia (17) and 46 new sequence types from non-
66 cultivated soils across Australia (18). These novel sequence types suggest that focusing only on
67 plant pathogenic *F. oxysporum* may underestimate the total species diversity. Other studies
68 indicate that *F. oxysporum* populations from soil are geographically heterogeneous, and greatly
69 influenced by resident plant populations (19, 20, 21), suggesting that some plants may exert a
70 selective effect on soil populations of *F. oxysporum* (22). Studies comparing pathogenic and
71 nonpathogenic *F. oxysporum* populations associated with the same host have found much
72 higher levels of diversity in the nonpathogenic populations than in the pathogens (11, 14, 23,
73 24).

74 An additional unaddressed topic is the diversity of *F. oxysporum* within an individual
75 plant. Endophytic *F. oxysporum* are commonly isolated from plants infected with pathogenic *F.*
76 *oxysporum* (for example, [23, 25]), suggesting that colonization of plants by multiple genotypes
77 of *F. oxysporum* may be common. Also, different genotypes may interact with their host plant
78 in different ways; for example, studies focusing on developing nonpathogenic *F. oxysporum* as
79 biocontrol agents have found particular isolates to be better biocontrol agents than others (26,
80 27, 28). A few studies have reported an over-representation of a few genotypes within
81 endophytic populations (12, 22), suggesting a possible selective effect by the host on soil
82 populations.

83 This study investigated the population structure of *F. oxysporum* associated with
84 asymptomatic tomato plants grown under field conditions. We hypothesized that endophyte
85 populations would be genetically differentiated from soil populations. Additionally, if only
86 specific genotypes of *F. oxysporum* can colonize plants endophytically, we expected that
87 endophyte populations would be less diverse than soil populations. The objective of the study
88 was to compare the population composition and diversity of *F. oxysporum* populations to look
89 for evidence of specialization of *F. oxysporum* to different environmental niches, such as soil or
90 plants. A further goal was to assess the diversity of *F. oxysporum* found within individual plants
91 and associated with different plant parts, namely roots, crowns, and stems.

92

93 **Material and methods**

94 *Sampling and fungal isolation*

95 *F. oxysporum* populations were sampled from two tomato fields located in the Russell E.
96 Larson Research Center, The Pennsylvania State University, PA Furnace, PA. These are
97 experimental fields used for agricultural research and are located approximately 1.5 km apart
98 from each other; they were chosen for sampling of *F. oxysporum* populations because they had
99 no known previous history of Fusarium wilt of tomato. Within each field, four plots were
100 randomly chosen for sampling of tomato plants and soil. Each plot consisted of four plants in a
101 row with one-foot spacing. Within a plot, six 4-inch soil cores were collected from between
102 each set of four plants and bulked together to make a composite soil sample per plot, for a total
103 of 16 plant samples and 4 soil samples per field. Plants in field 1 were all of the processing
104 tomato cultivar 'Heinz 9907'. In field 2, each set of four plants consisted of a different cultivar,
105 the processing tomato cultivars 'Gem 611' and 'Heinz 3402' and the fresh market cultivars 'FL
106 47' and 'Mountain Fresh'. These cultivars are all resistant to *F. oxysporum* f. sp. *lycopersici* races
107 1 and 2. No fungicides had been applied to the plants. Plants in field 1 were sampled at the
108 green fruit stage and plants in field 2 were sampled after the fruit were mature.

109 Sampling from plants was done by carefully digging up the plants, washing off the soil
110 under tap water, and surface-disinfesting entire plants with a solution of 2% commercial bleach
111 (6.15% NaOCl) for 1.5 minutes. Small (approximately 5 mm long) pieces of the root and stem
112 were placed on Nash-Snyder *Fusarium*-selective growth medium for fungal isolation (29). Soil
113 samples were air-dried for at least ten days and sieved to remove large particles. Five grams of
114 soil were suspended in 50 ml sterile water by mixing for 20 minutes, and then ten-fold serial
115 dilutions of 10^{-1} to 10^{-4} were plated on Nash-Snyder medium (24). A maximum of 10 isolates

116 were collected per plant: four from the roots and stem respectively and two from the crown,
117 for a maximum total of 40 isolates for each set of four plants (one plot); and a maximum of 40
118 isolates were collected per composite soil sample (one plot), selected randomly from dilution
119 plates. *F. oxysporum* isolates were identified based on colony morphology and monoconidial
120 cultures were grown on potato dextrose agar. Isolates were stored long-term at 4°C in 1.5 ml
121 microcentrifuge tubes in sterile sand mixed with 250 µl of potato dextrose broth.

122

123 *DNA extraction and sequencing*

124 DNA was extracted following the protocol of Ceniz (30). Briefly, mycelium was grown in
125 potato dextrose broth in 1.5 ml microcentrifuge tubes for at least three days at room
126 temperature, centrifuged to remove the medium, and ground in an SDS-based extraction
127 buffer. Cellular debris was precipitated with sodium acetate, and finally DNA was precipitated
128 with isopropanol. The translation elongation factor 1- α gene (TEF), a region commonly used for
129 *Fusarium* identification, was amplified using the primers EF1 and EF2 using published conditions
130 (16). Amplification was checked on agarose gels stained with ethidium bromide, and PCR
131 products were cleaned with EXO-SAP (Affymetrix, Santa Clara, CA) before sequencing.
132 Sequencing was done using the primers EF1 and EF2 at the Penn State Genomics Core Facility
133 (University Park, PA).

134

135 *Data analysis*

Sequences were compared to reference sequences in GenBank and the Fusarium-ID database (31) to confirm the isolates' identity as *F. oxysporum* and to identify if the sequences were present in the Fusarium-ID database or if they represent new sequence types. Isolates that were not *F. oxysporum* (5% of the total) were excluded from further analysis. Sequence types were defined and randomly numbered for each allele, including gaps as informative characters. Population size was estimated using the non-parametric estimator Chao 2, calculated in EstimateS 8.2.0 (R. K. Colwell, 2006 [purl.oclc.org/estimates]). Population genetic diversity was measured with Shannon's diversity index ($^S H_A$) based on the frequency of sequence types, using the program GENALEX 6 (32). Diversity was compared between subpopulations based on Shannon's mutual information index ($^S H_{UA}$) and significance was assessed using the log-likelihood test statistic G (33). Genetic differentiation among subpopulations was measured using two analyses: the nearest-neighbor statistic (S_{nn}) (34), which is based on nucleotide sequence and determines how often isolates with the most similar DNA sequences (the nearest neighbors) are found in the same population; and the haplotype statistic (H_{ST}), which is based on allele frequency: specifically $H_{ST} = 1 - (\text{average haplotype diversity in the subpopulations} / \text{average haplotype diversity in the total population})$ (35). Analyses were performed using DnaSP 5.10 (36), and statistical significance was assessed with 1,000 permutations using Monte Carlo simulations (35). Gaps were considered as a fifth character for H_{ST} analysis but not for S_{nn} . Sequences from different locations (i.e., plants or soil from each of the eight plots) were compared using principal coordinates analysis to look for clustering of sequences from particular environments. The principal coordinates analysis was

157 performed using UniFrac (37), with sequences weighted for abundance and non-normalized. A
158 phylogenetic network of the sequence types was estimated with statistical parsimony analysis
159 and was performed using TCS v. 1.21 (38). Maximum parsimony analysis was inferred from the
160 DNA sequences using the program PAUP* v.4.0b10 (D. L. Swofford, 2002, Sinauer Associates,
161 Sunderland, MA). Parsimony analysis was performed using the heuristic search option with the
162 MulTrees option on, 1,000 random addition sequences, the tree-bisection-reconnection
163 branch-swapping algorithm, and gaps not considered as characters. Support for each branch
164 was assessed with 1,000 bootstrap replicates. Included in the phylogenetic analysis were
165 sequences representative of the known diversity in *F. oxysporum* based on previous
166 phylogenetic analyses (16, 39, 40), downloaded from GenBank or the Fusarium-ID database
167 (31).

168

169 **Results**

170 *Fungal isolation*

171 *F. oxysporum* was successfully isolated from all eight soil samples and all 32 plants from
172 both fields, with a total of 609 isolates collected and analyzed: 314 isolates from soil and 295
173 from plants. A total of 16 populations were analyzed, eight from each field with four plots per
174 field. From each plot we established one population from plants and one from the
175 corresponding soil. For this research we have used a very broad definition of population as
176 group of individuals of the same species occupying a particular space or niche (in our case, a
177 plant or a soil) at a particular time (41). Isolates from plants were considered to be endophytes

178 because the plants were asymptomatic and surface disinfested. A maximum of ten isolates
179 were collected for each plant, and the majority of the plants yielded nine or ten isolates. The
180 plant which yielded the fewest isolates (six) was heavily colonized by *Geotrichum* spp. Soil
181 samples contained approximately 1,000 cfu of *F. oxysporum* per gram soil, based on
182 estimations from soil dilution plating.

183 Within each plant, *F. oxysporum* was isolated from the roots, crown, and stem of all
184 plants, although some plants yielded only one isolate per plant part. *F. oxysporum* was isolated
185 from the top third of the stem (approximately 60 cm above the soil) for 19 of the 32 plants,
186 suggesting that colonization of the entire stem is not uncommon. Whether *F. oxysporum* can
187 also be found in the leaves and fruit was not assessed.

188 *F. oxysporum* was the most common *Fusarium* spp. observed. Other *Fusarium* species
189 observed but excluded from further study (5% of the samples) were identified using the
190 Fusarium-ID database (31) as *F. commune*, *F. proliferatum*, and members of the *F. solani* species
191 complex, the *Gibberella fujikuroi* species complex, and the *F. incarnatum* – *F. equiseti* species
192 complex. *F. commune* was frequently isolated, but this was due to difficulties distinguishing it
193 morphologically from *F. oxysporum* and not to its relative abundance in the samples.

194

195 *Sequence types and population diversity*

196 A fragment of approximately 630 bp of the TEF gene was amplified for all isolates. Fifty-
197 seven sites were polymorphic, with 45 single nucleotide polymorphisms and 12
198 insertions/deletions (indels). Including indels as informative characters, 26 different sequence

199 types (STs) were observed (GenBank accession numbers KJ920404-KJ920429). Species
200 accumulation curves of populations from plants, soil, and plants and soil combined show that
201 sampling was approaching but did not reach saturation (Figure 1). The total population richness
202 was estimated to be 41 STs based on the Chao 2 estimator, with a 95% confidence interval from
203 29 to 88 STs. The Chao 2 richness estimate for soil was 21 STs, compared to 17 observed STs,
204 and 40 STs in plants, compared to 20 observed STs. Twenty-one of the 26 STs were found in
205 field 1, and 18 of the 26 STs were found in field 2. Two STs, 5 and 13, were especially common
206 and composed 35% and 30% of all isolates respectively. These two STs were equally common in
207 soil as in plant tissues, and were found in both fields, although ST 5 was twice as common in
208 field 2 compared to field 1 and ST 13 was three times more common in field 1 than in field 2.
209 The most abundant STs were found in both locations and in both soil and endophyte
210 populations, but some of the rare STs, including ten that were only observed once, were unique
211 to field 1 (eight STs), field 2 (five STs), plant tissues (eight STs only in roots or crowns, 1 ST only
212 in stems), or soil (six STs). Of the average of 9.2 isolates per plant, an average of 4.2 STs was
213 found per plant, ranging from two to seven. One to four STs were found per stem and per root
214 system, with an average of 2.2 STs per stem (out of an average of 3.5 isolates per stem) and 2.6
215 STs per roots (out of an average of 3.8 isolates per root system).

216 The diversity of populations from plants and from soil was compared using the
217 Shannon's diversity index (Table 1). Based on this index, endophytic populations were
218 significantly more diverse than soil populations for field 2 and for both fields combined. The
219 diversity was not significantly different between plant and soil populations for individual plots

220 except for plot 6 in field 2, in which plant populations were significantly more diverse than soil
221 populations.

222 Sequences were compared to the Fusarium-ID database to determine if these sequence
223 types had been found in previous samplings. Only 12 of the 26 sequence types had a 100%
224 match to sequences in the database. The two most common sequence types, ST 5 and ST 13,
225 have been previously found, associated with many different plant species. Most of the 14 novel
226 sequences types were rare, with only one or two isolates of each, although two of the novel
227 sequence types, ST 4 and ST 8, were found 44 and 32 times respectively.

228

229 *Genetic differentiation among populations*

230 The genetic differentiation between various subpopulations was calculated based on
231 both haplotypes (H_{ST}) and nucleotide sequence (S_{nn}) (Table 2). Gaps in the sequence alignment
232 were considered in calculations based on haplotypes, but results were similar when gaps were
233 excluded (results not shown). Populations were compared between the two fields to determine
234 if these were differentiated based on physical separation. Total populations were significantly
235 different between the two fields, which were approximately 1.5 km apart ($p < 0.001$ for both
236 S_{nn} and H_{ST}), as were the soil and endophyte populations in each field ($p < 0.001$ for both S_{nn}
237 and H_{ST}). Populations from plots in field 1, all planted with the same tomato cultivar, were
238 compared to determine if populations are also differentiated among locations several meters
239 apart. Based on S_{nn} , endophyte populations were significantly different for plot 1 compared to
240 plots 2 and 3, but not in any other pair-wise comparison. Soil populations were not significantly

241 differentiated among the plots based on S_{nn} or H_{ST} . Physical separation therefore does not
242 appear to be a significant cause of population differentiation on a small scale (meters) but is
243 significant on a larger scale (kilometers).

244 Populations were compared among plots in field 2, planted with different tomato
245 cultivars, to determine if host genotype affects *F. oxysporum* populations. Different sequence
246 types predominated in each cultivar, most notably ST 5 composing 72% of the endophyte
247 population from 'Heinz 3402' compared to 47% in 'Gem 611', 27% in 'FL 47', and 21% in
248 'Mountain Fresh' (Figure 2). These differences were reflected in the genetic differentiation
249 analyses using H_{ST} , which is based on allele frequency. Endophyte populations from 'Heinz
250 3402' were significantly different from populations in 'Gem 611' and 'Mountain Fresh' (Table
251 2). However, based on S_{nn} , which incorporates genetic distance between alleles, 'Heinz 3402'
252 was only significantly differentiated from 'Mountain Fresh'. Differences in allele frequency
253 between 'Heinz 3402' and 'Gem 611' therefore may be based on closely-related clones. Soil
254 populations were also compared to determine if differences between cultivars are biased by
255 the physical separation among plots. In pair-wise comparisons, no soil populations were
256 significantly differentiated based on either S_{nn} or H_{ST} , suggesting that the genetic differentiation
257 observed among endophyte populations was due to host genotype rather than different *F.*
258 *oxysporum* populations in different locations in the field.

259 Populations were also compared between endophyte and soil populations to determine
260 if populations were differentiated based on environmental niche. The total soil population was
261 differentiated from the total endophyte population for both fields combined and for field 2

262 based on S_{nn} ($p < 0.001$), but were not differentiated in field 1. The differentiation observed
263 between the total soil population and the total endophyte population could be biased by the
264 combination of populations from different locations and different tomato cultivars, so soil and
265 endophyte populations were compared for each plot. Endophyte and soil populations were
266 significantly differentiated for plot 5, planted with 'Gem 611' ($p < 0.001$ for both S_{nn} and H_{ST}),
267 but not for any of the other plots.

268 A principal coordinate analysis was performed on the sequence types to look for
269 clustering of isolations from different substrates. The first coordinate explained 56.79% of the
270 variation, the second coordinate explained 22.40% of the variation, and the third coordinate
271 explained 12.49% of the variation. In the plot of the first and second coordinates (Figure 3), the
272 samples were clustered based on whether they were from field 1 or field 2, but there was no
273 clear pattern of clustering based on plot or substrate, either soil or plants, although samples
274 from soil tended to be less dispersed in the plot than samples from plants.

275

276 *Phylogenetic analysis*

277 A statistical parsimony network of the sequence types was inferred, and the sequence
278 types grouped into two networks (Figure 4). Three loops were present in the larger network,
279 indicating homoplastic relationships or possibly recombination between isolates. Sequence
280 types mainly from soil were frequently closely related to sequence types mainly from plants,
281 although one branch of the tree was composed entirely of endophytic isolates (STs 17, 14, and
282 15).

283 A phylogenetic analysis of the tomato-associated sequence types compared to other *F.*
284 *oxysporum* isolates showed that the isolates were highly diverse and belonged to many
285 different clades within *F. oxysporum* (Figure 5). While some STs were closely related, no clear
286 pattern was observed based on whether isolates were from plant tissues or soil, and isolates
287 from soil and plants often shared the same sequence type. The two most common STs, 5 and
288 13, were not closely related. Most STs, including ST 5 and ST 13, were not closely related to
289 isolates known to be pathogenic to tomato, *forma specialis lycopersici* races 1, 2, and 3 and
290 *forma specialis radialis-lycopersici*.

291 Discussion

292 The main objectives of this study were to investigate the genetic diversity of *F.*
293 *oxysporum* populations associated with soil and asymptomatic tomato plants and to look for
294 evidence of specialization to particular environmental niches among nonpathogenic *F.*
295 *oxysporum*. Over 600 isolates were collected from plants and soil samples from two fields, and
296 endophytic *F. oxysporum* were successfully isolated from all plants sampled.

297 Populations of *F. oxysporum* living saprophytically in soil or nonpathogenically in
298 asymptomatic plant roots have been previously reported to be highly diverse. Most studies of
299 *F. oxysporum* populations used vegetative compatibility groups (VCGs) as a genetic diversity
300 estimator and usually found nonpathogenic populations to comprise large numbers of VCGs
301 even within one field, with the majority of isolates belonging to single-isolate VCGs (13, 15, 23,
302 24, 42). Studies using various molecular markers have reported 25 mtDNA haplotypes among
303 197 isolates from soil and asymptomatic melon roots (11), 23 ribosomal intergenic spacer-

304 restriction fragment length polymorphism (IGS-RFLP) types among 129 isolates from soil and
305 asymptomatic carnation roots (24), and 46 TEF sequence types among 214 *F. oxysporum*
306 isolates from soils across Australia (18). Twenty-six TEF sequence types were found in the two
307 tomato fields, suggesting a comparable level of diversity to previous studies given the
308 differences in sampling strategy and markers used.

309 *F. oxysporum* populations between the two tomato fields for both soil and plant
310 samples were significantly different ($p < 0.001$ for both S_{nn} and H_{ST}) (Table 2), although the two
311 fields were only approximately 1.5 km apart. Populations in each field were not entirely
312 isolated, as half of the 26 total STs were found in both fields, but the distribution of sequence
313 types was heterogeneous between the two fields. Populations of *F. oxysporum* in each field
314 might differ due to factors such as management practices, previous planting history, or soil type
315 (21).

316 *F. oxysporum* populations were also surprisingly diverse within each plant, with an
317 average of 4.2 STs found per plant, including roots and stems. *F. oxysporum* was isolated from
318 the stems of all plants, a niche that has not been studied for *F. oxysporum*, as previous
319 population genetics studies of nonpathogenic *F. oxysporum* have focused on the roots or crown
320 of plants (11, 12, 13, 15, 23). These isolations from tomato show that endophytic *F. oxysporum*
321 also commonly colonize plant stems. Within an individual stem, the presence of multiple STs
322 was common, with an average of 2.2 STs per plant. The ability of multiple inoculated *F.*
323 *oxysporum* genotypes to infect the same plant has been previously reported, such as for two
324 pathogenic isolates in muskmelon (43), two endophytic isolates in carnation (15), and a

325 pathogen and an endophyte in tomato (44) and in carnation (28). In addition, isolation of
326 nonpathogenic *F. oxysporum* from plants also infected with pathogenic *F. oxysporum* is
327 common (for example, 25, 45). Endophytic *F. oxysporum* therefore do not necessarily exclude
328 other *F. oxysporum* from colonizing the same plant, and wild plants might commonly harbor a
329 population of *F. oxysporum* endophytes rather than just a single individual.

330 One unexpected result of this study was that endophytic populations were equally or
331 sometimes more genetically diverse than populations from the surrounding soil (Table 1, Figure
332 1). Our initial hypothesis was that endophytic populations would be less diverse than soil
333 populations if the interaction between plant and fungal endophyte is somewhat specific,
334 leading to selection by the plant acting as a filter on the resident fungal soil population. Instead,
335 Shannon's diversity index indicated that overall populations isolated from plants were
336 significantly more diverse than populations from soil ($p < 0.001$). In individual plots, the levels of
337 diversity were not significantly differentiated, except for plot 6, in which levels of diversity were
338 much higher for the endophyte population ($p < 0.001$). In the principal coordinate analysis
339 (Figure 3), isolates from the crown and root are the most widely spread out, while isolates from
340 the soil tend to cluster together. These results might be an artifact of the sampling scheme as
341 more volume of plant material was collected than soil, but calculations of the population sizes
342 based on the Chao 2 non-parametric richness estimator suggest that the populations were only
343 slightly under-sampled. One possible explanation is that STs found only in the plants could have
344 colonized the plants prior to transplantation, either by seed transmission or during the seedling
345 stage before planting. This hypothesis is supported by the higher degree of genetic

346 differentiation among plots for endophyte populations than for soil populations (Table 2),
347 suggesting that soil populations are relatively uniform across the field and not the source of the
348 unique endophytic genotypes. Given that some *F. oxysporum* obtained from soil have been
349 found to suppress disease by pathogenic *F. oxysporum* (46) but other endophytic *F. oxysporum*
350 may be pathogenic to plants species besides the endophyte host (10, 47), the introduction of
351 new genotypes of endophytic *F. oxysporum* conceivably could increase or decrease the
352 incidence of Fusarium wilt and other diseases and may have important implications for
353 agriculture.

354 No definitive evidence was found to support the hypothesis that certain genotypes of *F.*
355 *oxysporum* are specialized to colonize tomato plants as endophytes. If only specific genotypes
356 can form an endophytic association with tomato, we hypothesize that populations within the
357 plants would be genetically differentiated from populations in the soil. This was true across
358 both fields combined and for field 2 (based on S_{nn} analysis, $p < 0.001$) (Table 2). However, closer
359 analysis did not show consistent differentiation between populations from different parts of
360 the plants (roots and stems) and the soil or between soil and plant populations for each plot
361 within the fields (Table 2). Moreover, diversity was generally not statistically different for
362 populations from plants compared to populations from soil (Table 1). If only specialized
363 genotypes could colonize the plant, diversity within the plant should be lower than diversity in
364 the soil, as is seen for pathogenic isolates of *F. oxysporum* compared to soil populations (14).

365 One result that may suggest selection of specialized genotypes within the population is
366 the predominance of two STs, 5 and 13, in all samples. The majority of isolates belonged to ST 5

367 and ST 13, including isolates from both fields and from both plants and soil. ST 5 and ST 13 may
368 be better adapted to colonizing tomato plants, or perhaps plants in general, than other isolates,
369 leading these STs to prevail in endophytic populations, as was suggested for a population of *F.*
370 *oxysporum* isolated from asymptomatic celery roots in which most isolates belonged to one of
371 two VCGs (12). The high prevalence of these sequence types in the soil could be due to a
372 selective effect of the plant roots on soil populations, or higher populations in the plant may
373 lead to higher populations in the surrounding soil. Alternatively, the frequent isolation of ST 5
374 and ST 13 may reflect their abundance in the local *F. oxysporum* soil population and not a
375 selective effect by tomato. Soil was not sampled from fields not cultivated with tomato, so it
376 cannot be determined if STs 5 and 13 are equally common in other local fields in the absence of
377 tomato plants.

378 Based on haplotype network results, isolates were not necessarily more closely related
379 to other isolates from the same niche than to isolates from other substrates, whether soil or
380 the roots, crown, or stem of the plants (Figure 4). Likewise, in a phylogenetic analysis of the
381 isolates from tomato compared to *F. oxysporum* isolated from other hosts, genotypes
382 associated with asymptomatic tomato, either from plants or from soil, were not necessarily
383 closely related (Figure 5). Some sequence types differed by only one single-nucleotide
384 polymorphism and therefore clustered together in the phylogeny, while other sequence types
385 were quite different. These results agree with the findings of a large phylogenetic analysis of *F.*
386 *oxysporum* (16), in which the included putative nonpathogens from several host plants did not
387 belong to any specific lineage within the complex.

388 The potential effect of host genotype on population composition was studied using the
389 samples from field 2, where four different tomato cultivars were sampled. Populations of
390 endophytic isolates were significantly different among cultivars based on H_{ST} ($p < 0.0001$) (Table
391 2). Isolates from 'Heinz 3402' in particular were differentiated from 'Gem 611' and 'Mountain
392 Fresh' based on H_{ST} . No clear patterns were observed based on whether the cultivars were bred
393 for fresh market ('FL 47' and 'Mountain Fresh') or for processing ('Gem 611' and 'Heinz 3402').
394 The differentiation of endophyte populations among cultivars suggests that particular host
395 genotypes interact differently with *F. oxysporum* genotypes, leading to different *F. oxysporum*
396 populations within each cultivar. This interpretation is tenuous, however, as differentiation
397 among endophyte populations was also observed among the plots in field 1, which contained
398 only one cultivar, 'Heinz 9907', and an uneven distribution of STs in the field could influence the
399 results. However, 'Gem 611' and 'Heinz 3402' were planted next to each other and harbored
400 distinct populations, and no significant differentiation was seen among soil populations in field
401 2, which suggests that differences in endophyte populations are not due to uneven spatial
402 distribution of STs. While the patterns observed are suggestive of the host genotype influencing
403 endophyte populations, firm conclusions cannot be drawn from these results.

404 Overall, the high amount of diversity observed among these populations highlights the
405 importance of nonpathogenic lifestyles to the biology of *F. oxysporum*. Both soil and endophyte
406 populations were composed of many sequence types, even associated with just one plant.
407 Some results were suggestive of specialization of endophytic populations, but generally support
408 the hypothesis that most, if not all, *F. oxysporum* are capable of living as endophytes. These

409 endophytic and soil populations may represent a source of unstudied diversity within *F.*
410 *oxysporum*. *F. oxysporum* is well known for its variety of agriculturally-important characteristics,
411 such pathogenicity on many plant hosts, production of secondary metabolites including
412 mycotoxins, and biocontrol activity, and better exploring endophytic and soil environments
413 may discover even more economically or scientifically relevant traits.

414

415

416 **Acknowledgments**

417 This study was partially funded by a grant to JED through the Pennsylvania State University
418 College of Agricultural Sciences graduate students competitive grants program.

419 We thank Dr. Majid Foolad for graciously allowing us to sample in his tomato field and Ms.
420 Tiana Montgomery-Noel for technical assistance.

421

422 **References**

- 423 1. **Stoner MF.** 1981. Ecology of *Fusarium* in noncultivated soils, p 276-286. *In* Nelson PE,
424 Toussoun TA, Cook RJ (ed), *Fusarium: diseases, biology, and taxonomy*. The
425 Pennsylvania State University, University Park, PA.
- 426 2. **Kuldau GA, Yates IE.** 2000. Evidence for *Fusarium* endophytes in cultivated and wild
427 plants, p 85-117. *In* Bacon CW, White JF Jr (ed), *Microbial endophytes*. Marcel Dekker,
428 Inc., New York.
- 429 3. **Olivain C, Alabouvette C.** 1999. Process of tomato root colonization by a pathogenic
430 strain of *Fusarium oxysporum* f. sp. *lycopersici* in comparison with a non-pathogenic
431 strain. *New Phytol.* **141**:497-510.
- 432 4. **Olivain C, Trouvelot S, Binet MN, Cordier C, Pugin A, Alabouvette C.** 2003. Colonization
433 of flax roots and early physiological responses of flax cells inoculated with pathogenic
434 and nonpathogenic strains of *Fusarium oxysporum*. *Appl. Environ. Microbiol.* **69**:5453-
435 5462.

- 436 5. **Baayen RP.** 1988. Responses related to lignification and intravascular periderm
437 formation in carnations resistant to *Fusarium* wilt. *Can. J. Bot.* **66**:784-792.
- 438 6. **Beckman CH, Mueller WC, Tessier B J, Harrison NA.** 1982. Recognition and callose
439 deposition in response to vascular infection in *Fusarium* wilt-resistant or susceptible
440 tomato plants. *Physiol. Plant Pathol.* **20**:1-10.
- 441 7. **Harling R, Taylor GS.** 1985 A light microscope study of resistant and susceptible
442 carnations infected with *Fusarium oxysporum* f. sp. *dianthi*. *Can. J. Bot.* **63**:638-646.
- 443 8. **Benhamou N, Garand C.** 2001. Cytological analysis of defense-related mechanisms
444 induced in pea root tissues in response to colonization by nonpathogenic *Fusarium*
445 *oxysporum* Fo47. *Phytopathol.* **91**:730-740.
- 446 9. **Leoni C, de Vries, M., ter Braak CJF, van Bruggen AHC, Rossing WAH.** 2013. *Fusarium*
447 *oxysporum* f.sp. *cepae* dynamics: in-plant multiplication and crop sequence simulations.
448 *Eur. J. Plant Pathol.* **137**:545-561.
- 449 10. **Katan J.** 1971. Symptomless carriers of the tomato *Fusarium* wilt pathogen.
450 *Phytopathol.* **61**:1213-1217.
- 451 11. **Appel DJ, Gordon TR.** 1994. Local and regional variation in populations of *Fusarium*
452 *oxysporum* from agricultural field soils. *Phytopathol.* **84**:786-791.
- 453 12. **Correll JC, Puhalla JE, Schneider RW.** 1986. Vegetative compatibility groups among
454 nonpathogenic root-colonizing strains of *Fusarium oxysporum*. *Can. J. Bot.* **64**:2358-
455 2361.

- 456 13. **Elias KS, Schneider RW, Lear MM.** 1991. Analysis of vegetative compatibility groups in
 457 nonpathogenic populations of *Fusarium oxysporum* isolated from symptomless tomato
 458 roots. Can. J. Bot. **69**:2089-2094.
- 459 14. **Katan T, Katan J.** 1988. Vegetative compatibility grouping of *Fusarium oxysporum* f. sp.
 460 *vasinfectum* from tissue and the rhizosphere of cotton plants. Phytopathol. **78**:852-855.
- 461 15. **Katan T, Berliner R, Katan J.** 1994. Vegetative compatibility in populations of *Fusarium*
 462 *oxysporum* from wild carnation. Mycol. Res. **98**:1415-1418.
- 463 16. **O'Donnell K, Gueidan C, Sink S, Johnston PR, Crous PW, Glenn A, Riley R, Zitomer NC,**
 464 **Colyer P, Waalwijk C, Lee Tvd, Moretti A, Kang S, Kim H-S, Geiser DM, Juba JH, Baayen**
 465 **RP, Crome MG, Bithell S, Sutton DA, Skovgaard K, Ploetz R, Kistler HC, Elliott M, Davis**
 466 **M, Sarver BAJ.** 2009. A two-locus DNA sequence database for typing plant and human
 467 pathogens within the *Fusarium oxysporum* species complex. Fungal Genet. Biol. **46**:936-
 468 948.
- 469 17. **Balmas V, Migheli Q, Scherm B, Garau P, O'Donnell K, Ceccherelli G, Kang S, and Geiser**
 470 **DM.** 2010. Multilocus phylogenetics show high levels of endemic Fusaria inhabiting
 471 Sardinian soils (Tyrrhenian Islands). Mycologia **102**:803-812.
- 472 18. **Laurence MH, Burgess LW, Summerell BA, Liew ECY.** 2012. High levels of diversity in
 473 *Fusarium oxysporum* from non-cultivated ecosystems in Australia. Fungal Biol. **116**:289-
 474 297.

- 475 19. **Edel V, Steinberg C, Gautheron N, Recorbet G, Alabouvette C.** 2001. Genetic diversity
476 of *Fusarium oxysporum* populations isolated from different soils in France. FEMS
477 Microbiol. Ecol. **36**:61-71.
- 478 20. **Gordon TR, Okamoto D.** 1992. Variation within and between populations of *Fusarium*
479 *oxysporum* based on vegetative compatibility and mitochondrial DNA. Can. J. Bot.
480 **70**:1211-1217.
- 481 21. **Gordon TR, Okamoto D, Milgroom MG.** 1992. The structure and interrelationship of
482 fungal populations in native and cultivated soils. Mol. Ecol. **1**:241-249.
- 483 22. **Edel V, Steinberg C, Gautheron N, Alabouvette C.** 1997. Populations of nonpathogenic
484 *Fusarium oxysporum* associated with roots of four plant species compared to soilborne
485 populations. Phytopathol. **87**:693-697.
- 486 23. **Alves-Santos FM, Benito EP, Eslava AP, Díaz-Mínguez JM.** 1999. Genetic diversity of
487 *Fusarium oxysporum* strains from common bean fields in Spain. Appl. Environ.
488 Microbiol. **65**:3335-3340.
- 489 24. **Lori G, Edel-Hermann V, Gautheron N, Alabouvette C.** 2004. Genetic diversity of
490 pathogenic and nonpathogenic populations of *Fusarium oxysporum* isolated from
491 carnation fields in Argentina. Phytopathol. **94**:661-668.
- 492 25. **Fiely MB, Correll JC, Morelock TE.** 1995. Vegetative compatibility, pathogenicity, and
493 virulence diversity of *Fusarium oxysporum* recovered from spinach. Plant Dis. **79**:990-
494 993.

- 495 26. **Bao J, Fravel D, Lazarovits G, Chellemi D, van Berkum P, O'Neill N.** 2004. Biocontrol
496 genotypes of *Fusarium oxysporum* from tomato fields in Florida. *Phytoparasitica* **32**:9-
497 20.
- 498 27. **Larkin RP, Fravel DR.** 1998. Efficacy of various fungal and bacterial biocontrol organisms
499 for control of *Fusarium* wilt of tomato. *Plant Dis.* **82**:1022-1028.
- 500 28. **Postma J, Luttikholt AJG.** 1996. Colonization of carnation stems by a nonpathogenic
501 isolate of *Fusarium oxysporum* and its effect on *Fusarium oxysporum* f. sp. *dianthi*. *Can.*
502 *J. Bot.* **74**:1841-1851.
- 503 29. **Nash SM, Snyder WC.** 1962. Quantitative estimations by plate counts of propagules of
504 bean root rot *Fusarium* in field soils. *Phytopathol.* **52**:567-572.
- 505 30. **Cenis JL.** 1992. Rapid extraction of fungal DNA for PCR amplification. *Nuc. Acids Res.*
506 **20**:2380.
- 507 31. **Geiser DM, Jiménez-Gasco MM, Kang S, Makalowska I, Veeraraghavan N, Ward TJ,**
508 **Zhang N, Kuldau GA, O'Donnell K.** 2004. FUSARIUM-ID v. 1.0: A DNA sequence database
509 for identifying *Fusarium*. *Eur. J. Plant Pathol.* **110**:473-479.
- 510 32. **Peakall R, Smouse PE.** 2006. GENALEX 6: Genetic analysis in Excel. Population genetic
511 software for teaching and research. *Mol. Ecol. Notes* **6**:288-295.
- 512 33. **Sherwin WB, Jabot F, Rush R, Rossetto M.** 2006. Measurement of biological information
513 with applications from genes to landscapes. *Mol. Ecol.* **15**:2857-2869.
- 514 34. **Hudson RR.** 2000. A new statistic for detecting genetic differentiation. *Genetics*
515 **155**:2011-2014.

- 516 35. **Hudson RR, Boos DB, Kaplan NL.** 1992. A statistical test for detecting geographic
517 subdivision. *Mol. Biol. Evol.* **9**:138-151.
- 518 36. **Librado P, Rozas J.** 2009. DnaSP v5: A software for comprehensive analysis of DNA
519 polymorphism data. *Bioinformatics* **25**:1451-1452.
- 520 37. **Lozupone C, Hamady M, Knight R.** 2006. UniFrac – an online tool for comparing
521 microbial community diversity in a phylogenetic context. *BMC Bioinformatics* **7**:371.
- 522 38. **Clement M, Posada D, Crandall K.** 2000. TCS: a computer program to estimate gene
523 genealogies. *Mol. Ecol.* **9**:1657-1660.
- 524 39. **Baayen RP, O'Donnell K, Bonants PJM, Cigelnik E, Kroon LPNM, Roebroek EJA,**
525 **Waalwijk C.** 2000. Gene genealogies and AFLP analyses in the *Fusarium oxysporum*
526 complex identify monophyletic and nonmonophyletic formae speciales causing wilt and
527 rot disease. *Phytopathol.* **90**:891-900.
- 528 40. **O'Donnell K, Kistler HC, Cigelnik E, Ploetz RC.** 1998. Multiple evolutionary origins of the
529 fungus causing Panama disease of banana: Concordant evidence from nuclear and
530 mitochondrial gene genealogies. *Proc. Nat. Acad. Sci. USA.* **95**:2044-2049.
- 531 41. **Krebs CJ.** 1972. *Ecology: The Experimental Analysis of Distribution and Analysis.* Harper
532 and Row, New York, NY.
- 533 42. **Appel DJ, Gordon TR.** 1996. Relationships among pathogenic and nonpathogenic
534 isolates of *Fusarium oxysporum* based on the partial sequence of the intergenic spacer
535 region of the ribosomal DNA. *Mol. Plant-Microbe Interact.* **9**:125-138.

- 536 43. **Meyer JA, Maraite H.** 1971. Multiple infection and symptom mitigation in vascular wilt
537 diseases. *Trans. Brit. Mycol. Soc.* **57**:371-377.
- 538 44. **Olivain C, Humbert C, Nahalkova J, Fatehi J, L'Haridon F, Alabouvette C.** 2006.
539 Colonization of tomato root by pathogenic and nonpathogenic *Fusarium oxysporum*
540 strains inoculated together and separately into the soil. *Appl. Environ. Microbiol.*
541 **72**:1523-1531.
- 542 45. **Skovgaard K, Bødker L, Rosendahl S.** 2002. Population structure and pathogenicity of
543 members of the *Fusarium oxysporum* complex isolated from soil and root necrosis of
544 pea (*Pisum sativum* L.). *FEMS Microbiol. Ecol.* **42**:367-374.
- 545 46. **Alabouvette C.** 1990. Biological control of Fusarium wilt pathogens in suppressive soils,
546 p 27-43. *In* Hornby D (ed), *Biological control of soilborne plant pathogens*. CAB
547 International, Wallingford, UK.
- 548 47. **Helbig JB, Carroll RB.** 1984. Dicotyledonous weeds as a source of *Fusarium oxysporum*
549 pathogenic on soybean. *Plant Dis.* **68**:694-696.

550 **Table 1** Pair-wise comparisons of Shannon's diversity indices (sH_A) between populations from soil and plants.

	Plant population sH_A	Soil population sH_A	${}^sH_{UA}$ ^a	G ^b	P-value ^c
All locations	2.729	2.672	0.081	68.437	< 0.001
Field 1	2.607	2.440	0.077	33.354	0.031
Field 2	2.529	2.364	0.147	60.822	< 0.001
Plot 1 ('Heinz 9907')	2.318	2.478	0.150	16.044	0.098
Plot 2 ('Heinz 9907')	1.653	2.084	0.117	13.001	0.224
Plot 3 ('Heinz 9907')	2.709	1.703	0.225	24.041	0.020
Plot 4 ('Heinz 9907')	2.436	2.512	0.185	19.702	0.073
Plot 5 ('FL 47')	2.542	2.190	0.180	18.484	0.047
Plot 6 ('Gem 611')	2.710	1.619	0.351	37.426	< 0.001
Plot 7 ('Heinz 3402')	1.386	2.052	0.158	16.405	0.022
Plot 8 ('Mountain Fresh')	2.277	2.748	0.218	21.746	0.026

551 ^a ${}^sH_{UA}$ = Shannon's mutual information index

552 G = log-likelihood test statistic

553 ^c P-values of the log-likelihood test of differences between soil and endophyte populations, with statistically significant values

554 in bold [Bonferroni adjusted $p < 0.0045$, (0.05/11)].

555

Table 2 Genetic differentiation among *Fusarium oxysporum* subpopulations from different fields, plots and substrates.

Sampling location ^a	S_{nn} ^b	P	H_{ST} ^b	P
All soil isolates (314) vs. all plants isolates (295)	0.52244	< 0.001	0.00292	0.039
Field 1 (311) vs. field 2 (298)	0.59486	< 0.001	0.05196	< 0.001
Soil in field 1 (158) vs. soil in field 2 (156)	0.63467	< 0.001	0.07442	< 0.001
Plants in field 1 (153) vs. plants in field 2 (142)	0.55456	< 0.001	0.03237	< 0.001
Field 1				
Soil (158) vs. plants (153)	0.51963	0.011	-0.00064	0.5140
Soil (158) vs. roots (61)	0.61855	0.016	0.00134	0.1930
Soil (158) vs. stems (61)	0.60957	0.1050	-0.00169	0.6130
Roots (61) vs. stems (61)	0.52225	0.0690	0.00364	0.2000
Plot 1: soil (39) vs. plants (38)	0.53453	0.0820	0.00000	0.3740
Plot 2: soil (40) vs. plants (40)	0.48901	0.5250	0.00354	0.2630
Plot 3: soil (39) vs. plants (38)	0.52804	0.0930	0.03640	0.005
Plot 4: soil (40) vs. plants (37)	0.54687	0.037	0.00360	0.2490
By plots (plants only)	0.29965	< 0.001	0.04364	0.001
Plot 1 (38) vs. Plot 2 (40)	0.61930	< 0.001	0.08801	< 0.001
Plot 1 (38) vs. Plot 3 (38)	0.59271	0.001	0.01693	0.044
Plot 1 (38) vs. Plot 4 (37)	0.57874	0.004	0.03091	0.014
Plot 2 (40) vs. Plot 3 (38)	0.51912	0.1230	0.03503	0.008

Plot 2 (40) vs. Plot 4 (37)	0.49443	0.4640	0.01174	0.1020
Plot 3 (38) vs. Plot 4 (37)	0.48607	0.6080	-0.00411	0.6420
By plots (soil only)	0.25322	0.2640	0.02106	0.029
Plot 1 (39) vs. Plot 2 (40)	0.51622	0.1570	0.01159	0.1170
Plot 1 (39) vs. Plot 3 (39)	0.53279	0.0860	0.05149	0.003
Plot 1 (39) vs. Plot 4 (40)	0.47066	0.8000	-0.00825	0.9010
Plot 2 (40) vs. Plot 3 (39)	0.48180	0.6470	0.00149	0.2910
Plot 2 (40) vs. Plot 4 (40)	0.51827	0.1790	0.00086	0.3270
Plot 3 (39) vs. Plot 4 (40)	0.52057	0.1310	0.03139	0.020
Field 2				
Soil (156) vs. plants (142)	0.52643	< 0.001	0.01069	0.007
Plot 5 – ‘Gem 611’: soil (40) vs. plants (37)	0.62040	< 0.001	0.08284	< 0.001
Plot 6 – ‘FL 47’: soil (38) vs. plants (36)	0.50618	0.2780	-0.00176	0.4900
Plot 7 – ‘Heinz 3402’: soil (39) vs. plants (36)	0.48576	0.5740	0.01350	0.0920
Plot 8 – ‘Mountain Fresh’: soil (39) vs. plants (33)	0.51499	0.2470	0.01975	0.044
By cultivar (plants only)	0.28595	0.006	0.06590	< 0.001
‘Gem 611’ (37) vs. ‘FL 47’ (36)	0.50582	0.3120	0.00977	0.1190
‘Gem 611’ (37) vs. ‘Heinz 3402’ (36)	0.57121	0.003	0.07524	< 0.001
‘Gem 611’ (37) vs. ‘Mountain Fresh’ (32)	0.47132	0.7830	0.00284	0.3060
‘FL 47’ (36) vs. ‘Heinz 3402’ (36)	0.51714	0.1860	0.01999	0.048
‘FL 47’ (36) vs. ‘Mountain Fresh’ (32)	0.52684	0.1540	0.03992	0.008

'Heinz 3402' (36) vs. 'Mountain Fresh' (32)	0.61245	< 0.001	0.13219	< 0.001
By plots (soil only)	0.26894	0.025	0.01992	0.020
Plot 5 – 'Gem 611' (40) vs. Plot 6 – 'FL 47' (38)	0.56327	0.008	0.02852	0.020
Plot 5 – 'Gem 611' (40) vs. Plot 7 – 'Heinz 3402' (39)	0.50211	0.3490	0.00425	0.2150
Plot 5 – 'Gem 611' (40) vs. Plot 8 – 'Mountain Fresh' (39)	0.55374	0.022	0.04594	0.002
Plot 6 – 'FL 47' (38) vs. Plot 7 – 'Heinz 3402' (39)	0.49057	0.4670	-0.00354	0.5510
Plot 6 – 'FL 47' (38) vs. Plot 8 – 'Mountain Fresh' (39)	0.51342	0.2180	-0.00240	0.5500
Plot 7 – 'Heinz 3402' (39) vs. Plot 8 – 'Mountain Fresh' (39)	0.51798	0.1570	0.00912	0.1120

^a The number of samples from each sampling location is given in parentheses.

^b S_{nn} is the nearest-neighbor statistic and H_{ST} is the haplotype statistic. Statistically significant values are in bold [Bonferroni adjusted $p < 0.001$, (0.05/45)]

567 **List of figures**

568 **Fig. 1** Species accumulation curves showing the expected number of sequence types when
569 sampling a given number of individuals. Curves show the observed number of sequence types
570 for samples from plants and soil combined (squares), plants only (triangles), and soil only
571 (circles). The Chao 2 non-parametric richness estimator, calculated in EstimateS 8.2.0 (Colwell
572 2006), is also shown for samples from plants and soil combined (diamonds).

573

574 **Fig. 2** Comparison of sequence type frequencies among populations. The frequency of each
575 sequence type is shown for all plots in field 1 (153 isolates from plants, 158 isolates from soil)
576 and for each cultivar in field 2: 'FL 47' (36 isolates from plants, 38 isolates from soil), 'Gem 611'
577 (37 isolates from plants, 40 isolates from soil), 'Heinz 3402' (36 isolates from plants, 39 isolates
578 from soil), and 'Mountain Fresh' (33 isolates from plants, 39 isolates from soil). Sequence type
579 frequencies were calculated using GENALEX 6 (Peakall & Smouse 2006).

580

581 **Fig. 3** Principal coordinates analysis of sequences from different plots and substrates. Symbols
582 represent samples from different substrates (circles: from soil, squares: from root, diamonds:
583 from crown, triangles: from stem). Black symbols represent samples from field 1, and white
584 symbols represent samples from field 2. Sequences were weighted for abundance and the
585 analysis was not normalized.

586

587 **Fig. 4** Statistical parsimony network of the translation elongation factor 1- α sequences types,
588 constructed in TCS v. 1.21 (Clement *et al.* 2000). Sequence types are represented by circles,
589 with the number given to each sequence type in the center of the circle, and the size of the
590 circle proportional to the number of isolates with that sequence type. Grayscale represent the
591 percentage of isolates in each sequence type from different substrates (black – soil, dark gray –
592 roots, light gray – crowns, white – stems). The inferred ancestral sequence types are indicated
593 with stars. Branch lengths are not to scale.

594
595 **Fig. 5** One of two most parsimonious trees based on the translation elongation factor 1- α gene
596 using maximum parsimony analysis (81 steps, consistency index = 0.926, retention index =
597 0.968, rescaled consistency index = 0.897). The tree is rooted with *F. commune* NRRL 38348 and
598 bootstrap support >50% based on 1,000 replicates is indicated for each branch. Sequence types
599 generated in this study from tomato fields are in bold, with the number of isolates per
600 sequence type and the original substrates (plants or soil) given in parentheses. Arrows indicate
601 the two most abundance sequences types, ST 5 and ST 13.

602









